2.4.2 Theorem (Hopf-Rinow). *Let (M, g) be a connected RMF, then the following conditions are equivalent:*

(MC) The metric space (M, d) is complete (i.e., every Cauchy sequence converges).

(GC') There is $p \in M$ such that M is geodesically complete at p, i.e., $\exp_p$ is defined on all of T_pM.

(GC) (M, g) is geodesically complete.

(HB) M possesses the Heine-Borel property, i.e., every closed and bounded subset of M is compact.